



## AQ2.6: Activity Questions 6 - Not Graded



This assignment will not be graded and is only for practice.

**Level 1:**

*1 point*

Let  $Ax = b$  be a matrix representation of a system of linear equations. Let  $[R|c]$  be the reduced row echelon of the augmented matrix  $[A|b]$  corresponding to the system. Choose the set of correct options.

[Hint: Recall reduced row echelon form.]

☐

If the system  $Rx = c$  has infinitely many solutions, then the system  $Ax = b$  has infinite solutions.

☐

If the system  $Rx = c$  has no solutions, then the system  $Ax = b$  has a unique solution.

☐

If the system  $Rx = c$  has a unique solution, then the system  $Ax = b$  has no solution.

☐

If the system  $Rx = c$  has a unique solution, then the system  $Ax = b$  has a unique solution.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

If the system  $Rx = c$  has infinitely many solutions, then the system  $Ax = b$  has infinite solutions.

If the system  $Rx = c$  has a unique solution, then the system  $Ax = b$  has a unique solution.

Consider a system of linear equations

$$\begin{aligned} 2x_1 + x_2 &= 1 \\ -x_1 + x_3 + x_4 &= -1 \\ x_1 + x_2 - x_3 + x_4 &= 2 \\ -x_1 + x_3 + x_4 &= 1. \end{aligned}$$

$-x_3 + x_4 = 2 - x_1 + x_3 + x_4 = 1$ . If the following matrix represents the augmented matrix of the system, then answer the questions 2, 3 and 4.

Find the value of  $a_{22}$ .

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 0

*1 point*

Find the sum of the elements of the 4-th row of the augmented matrix.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**



(Type: Numeric) 2

1 point

Find the sum of the elements of the 3-rd column of the augmented matrix.

**No, the answer is incorrect.****Score: 0****Accepted Answers:**

(Type: Numeric) 1

1 point

1 point

Let  $Ax = b$  be a matrix representation of a system of linear equations and  $b = 0$ . Choose the set of correct options.

[Hint: Recall the definition of the trivial solution of a system of linear equations and applicability of Gauss Elimination Method.]

☐

If  $AA$  is an invertible matrix then the system has no solution.

☐

If  $AA$  is an invertible matrix then the system has a unique solution.

☐

If  $AA$  is an invertible matrix then the trivial solution is the only solution for the system.

☐

If  $\det(A) = 0$ , then the system has infinitely many solutions.

**No, the answer is incorrect.****Score: 0****Accepted Answers:**

If  $AA$  is an invertible matrix then the system has a unique solution.

If  $AA$  is an invertible matrix then the trivial solution is the only solution for the system.

If  $\det(A) = 0$ , then the system has infinitely many solutions.

1 point

Let  $Ax = b$  be a matrix representation of a system of linear equations and  $b \neq 0$ . Choose the set of correct options.

☐

If  $AA$  is an invertible matrix, then the system has no solution.

☐

If  $AA$  is an invertible matrix, then the system has a unique solution.

☐

If  $AA$  is an invertible matrix, then the trivial solution is a solution for the system.

☐

If  $\det(A) = 0$ , then either the system has no solution or the system has infinitely many solutions.

**No, the answer is incorrect.****Score: 0****Accepted Answers:**

If  $AA$  is an invertible matrix, then the system has a unique solution.

If  $\det(A) = 0$ , then either the system has no solution or the system has infinitely many solutions.

1 point

Consider the following systems of linear equations:

$$\begin{aligned} \text{System I: } & x - y = 3 \\ & -y + 2z = 1 \\ & x + y + z = 0 \end{aligned}$$

$$\begin{aligned} \text{System II: } & x - y = 3 \\ & 2y + z = 1 \end{aligned} \quad \text{System I: System II: System III:}$$

$$\begin{aligned} \text{System III: } & x - y = 3 \\ & 2y + z = 1 \\ & 6y + 3z = 3 \end{aligned}$$

$$x - y = 3 \quad -y + 2z = 1 \quad x + y + z = 0 \quad x - y = 3 \quad 2y + z = 1 \quad 6y + 3z = 3$$

Choose the correct option.

- ☐ All the three systems have a unique solution.
- ☐ System I has a unique solution.
- ☐ System II has no solution.
- ☐ System III has infinitely many solutions.
- ☐ Both the System II and III have infinitely many solutions.
- ☐ Both the system II and III have no solution.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

System I has a unique solution.

System II has no solution.

System III has infinitely many solutions.

**Level 2:**

Suppose  $P$  is a  $3 \times 3$  real matrix as follows:  $P = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

The vector  $X_n$  is defined by the recurrence relation  $P X_{n-1} = X_n$ . If  $X_2 = \begin{bmatrix} 7 \\ 8 \\ 3 \end{bmatrix}$ ,

$X_0 = \begin{bmatrix} \quad \\ 7 \\ 8 \\ 3 \end{bmatrix}$ , what is the sum of all the elements of  $X_0$ ?

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 6

1 point

1 point

Consider the system of linear equations given below:

$$\begin{aligned} x - y + z &= 2 \\ x + y - z &= 3 \\ -x + y + z &= 4. \end{aligned}$$

The system of linear equations has

[Hint: Use the relation between a system of linear equations and determinant of corresponding coefficient matrix.]

- ☐ no solution.  
☐ infinitely many solutions.  
☐ a unique solution.  
☐ finitely many solutions.  
☐ 3 dependent variable.  
☐ 2 dependent and 1 independent variable.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

a unique solution.

3 dependent variable.

1 point

Consider the system of linear equations:

$$\begin{aligned} -x_1 + x_2 + 2x_3 &= 1 \\ 2x_1 + x_2 - 2x_3 &= -1 \\ 3x_2 + cx_3 &= d \end{aligned}$$

Choose the set of correct options.

[Hint: Recall the Gauss elimination method.]

☐

If  $c = 2$  and  $d = 1$ , then the system has infinitely many solutions.

☐

If  $c = 1$  and  $d = 1$ , then the system has infinitely many solutions.

☐

If  $c = 1$  and  $d = 2$ , then the system has no solution.

☐

If  $c = 3$  and  $d = 2$ , then the system has a unique solution.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**